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Product Intelligence: Theory and Practice

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Abstract: Are there any benefits in allowing orders and products to be able to manage their own progress through a supply chain? The notion of associating (and even embedding) information management and reasoning capabilities with a physical product has been discussed for over ten years now. This talk will review the notions of product intelligence and examine the rationales for these models and the practicality of their implementation. Both theoretical and practical issues associated with product intelligence will be examined referencing a number of trial deployments in manufacturing, logistics and aerospace equipment servicing.

Keywords: intelligent product, pull control systems, distributed information system, distributed intelligence

1. INTRODUCTION

1.1 What is Product Intelligence?

In an industrial context, product intelligence is the phrase used to describe the linking of a physical order or product to information and rules governing the way it is intended to be made, stored or transported. The information and rules might for example represent the needs or intentions of the customer with regard to how the product might be managed by supplying organisations. Such a facility then enables the product – and hence indirectly the customer - to to be represented within a supply organisation's information systems environment. Figure 1 illustrates a production order generating an intelligent product to negotiate an assembly path with different production resources.

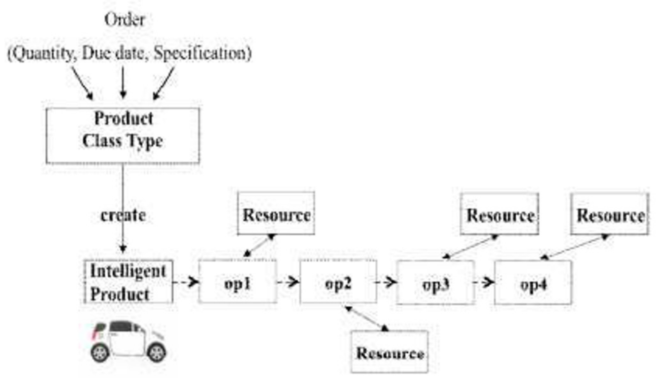

Fig. 1. Product Intelligence in a Manufacturing Context

1.2 Why Consider Product Intelligence for Industrial Systems?

Following from the previous section, one clear motivation for a product intelligence approach is to provide a customer with the opportunity to inform and influence the supply of its orders. *Although not explicitly mentioned yet, there is an assumption that this is done in an automated manner.*

In McFarlane, et al (2012) a number of situations were identified where the needs or interests of the order (or the order owner) might differ from those of the supplier handling the order. These were grouped into static conditions – those relating to the structure of the situation and dynamic conditions – those relating to operation variations:

Static Scenarios	*Dynamic Scenarios*
When a product or order moves between organizations in its delivery.	When options arise frequently and unpredictably for alternative routings to be considered.
When a specific item can be part of multiple orders/ consignments for certain stages of its production/ delivery.	When disruptions are frequent and performance guarantees are difficult to achieve.
When a customer's specific requirements for his order is at odds with the aggregate intentions of the logistics organisation.	When decision making about order management requires human resources that are not available.
When an order exists in multiple segments scattered across multiple organizations.	When customer's preferences change between ordering and delivering.
Where the different players in the logistics chain have only partial or no information sharing.	When an item's characteristics change over time.

Table 1. Product Intelligence Scenarios (McFarlane et al, 2012)

These scenarios highlight situations where the nature of the processing of the product might change or where the customer and supplier may have either different perspectives, priorities or where the supplier may have a limited ability to access information or influence the progress of an order. Hence we might refer to product intelligence as providing an *order or customer-oriented approach* as opposed to a more conventional *organisation-oriented approach* to managing operations. We will discuss this interpretation further in the next section. In this section we gather together various theoretical aspects associated with intelligent products.

2.1 Definitions & Features

There are numerous definitions of an inteilligent product. In general these describe the features of the intelligent product rather than what it means to be an intelligent product. We therefore include the following working definition:

Definition *1: Intelligent Product [Descriptive]*

A physical order or product that is linked to information and rules governing the way it is intended to be made, stored or transported that enables the product to support or influence these operations.

Several authors have described product intelligence in terms of the features it might essentially or optionally display (Kärkkäinen, M., et al. (2003), Wong et al, (2002)) for example:

Definition 2: Intelligent Product [Features] (Wong et al., 2002)

An Intelligent Product is a product (or part or order) that has part or all of the following five characteristics:

1. *Possesses a unique identity*
2. *Is capable of communicating effectively with its environment*
3. *Can retain or store data about itself*
4. *Deploys a language to display its features, production requirements etc.*
5. *Is capable of participating in or making decisions relevant to its own destiny*

Further product intelligence has also been described in terms of defined levels associated with that model as per Wong et al (2002) below or in Meyer et al (2009).

Definition 3: Product Intelligence [Levels] (Wong et al., 2002)

- *Level 1 Product Intelligence: which allows a product to communicate its status (form, composition, location, key features), i.e. it is information-oriented.*
- *Level 2 Product Intelligence: which allows a product to assess and influence its function (e.g. self-distributing inventory and self-manufacturing inventory) in addition to communicating its status, i.e. it is decision-oriented.*

We note that in other papers the conceptual internal architecture of an intelligent product is discussed (Valckenaers et al, 2008) in which the information and physical roles and an interface between them is outlined.

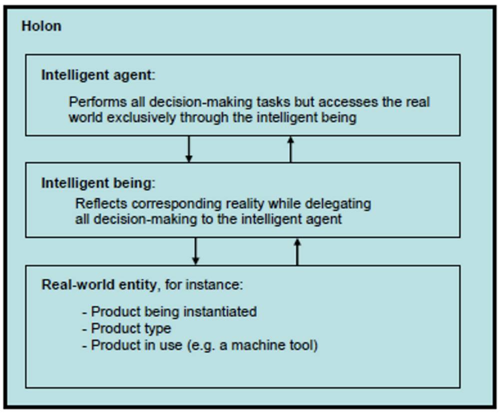

Fig. 2. Intelligent Product Structure (Valckenaers, 2008)

2.2 Characteristics/Interpretations of Product Intelligence

In Section 1 we introduced product intelligence as providing a framework for an *order or customer-oriented approach* as opposed to a more conventional *organisation-oriented approach* to managing operations.

The key characteristics of a system exhibiting product intelligence[1] is that the order or individual products can:

- Represent the (customer) needs linked to the order: e.g. goods required, quality, timing, cost agreed
- Communicate with the local organisation (as well as with the customer for the order)
- Monitor/track the progress of the order through the industrial supply chain
- [Using the preferences of the customer] influence the choice between different options affecting the order when such a choice needs to be made
- Adapt depending on conditions.

Industrially, we note that the customer-oriented view of intelligent product approach has resonances with the notion of *pull-systems* (see Slack, 2001) – in which downstream customer requirements *pull* an order through a series of operations. Mechanisms such as *kanbans* to drive supplier processes (see Fig 3) and the use of so called *product recipes* to specify product order requirements to their chemical suppliers (Convanich et al, 2008) as depicted in Figure 4.

[1] The notion of of an external customer or organisation seeking to represent its interests within the operations of another is not a new notion as the the idea is aligned with that of military envoys, government embassies, guided tour operators etc. What is less common is the automation of this representation.

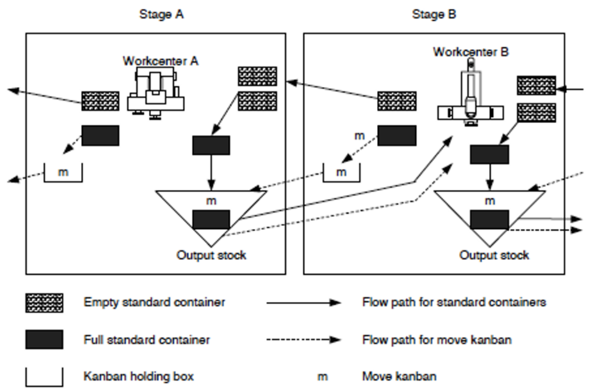

Fig 3: Kanban driven Operations (from Slack et al, 2001)

Further, in online consumer goods retailing a customer builds up an order specification which not only describes the items ordered but also customer preferences for delivery options and timings and replacement item selections. Each of these represents an example of the customer embedding some of its requirements into the operations of a supplier – and in some cases with the ability to influence those operations.

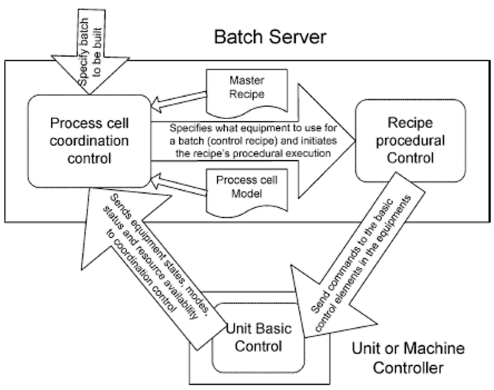

Fig 4 Master Recipe [from customer] driving production (source Covanich, 2008)

We finally note that distributed intelligent paradigms such as agent-based control (e.g. Paranuk, 1999, Marik, 2005) and holonic manufacturing control (e.g. Van Brussel, 1998, Deene, 2003) can provide a distributed framework for customer oriented or product intelligence driven operations as well as also supporting more conventionally organisationally oriented operations.

Hence we can summarise the essence of a control system based on a product intelligcnce approach as being one which is

a) distributed in terms of information and/or decision processing;
b) customer pull oriented; and
c) fully or partially automated.

2.3. Theoretical Challenges

In the next section we will discuss some of the challenges associated with deploying approaches based on product intelligence in an industrial context. But here we also briefly note that some theoretical challenges exist as well.

Design Challenges

These are challenges associated with the design and structure of the intelligent product environment:

- *Modelling Approaches:* Identification of modelling environments which support a) representation of the product, its information and decision environments b) representation of suitable the industrial environment in which the product operates c) representation of the existing planning and control systems.

- *Language*: Product intelligence implies that the customer order or product being able to communicate with resources or operations within multiple different organisations. This means either a) A common language for al related orders and resources or a means of interpreting between tasks required by orders and functionalities supplied by resources.

- *Interaction Architecture:* Development of system architectures that support the interaction between multiple orders/products and multiple resources across multiple organisations. A challenge is to be able to support this system in an efficient and scalable manner

Operational Challenges

These are challenges associated with the operation, behaviour and performance of the intelligent product environment:

- *Conflict Handling:* Enabling multiple intelligent products from different customers to cooexist in the same supply enviroment without creating

- *Disruption Management/Robustness:* Developing capabilities for intelligent products to adapt automatically to change or disruption conditions.

- *Performance:* Establishing appropriate measures for measuring a) the system performance of intelligent product based environments (which are inherently distributed) and b) the operational performance which is ultimately the measures that will influence end users of the approach.

History has shown that addressing theoretical challenges associated with a new approach to managing operations evolves in parallel with efforts to deploy the paradigm in practice. In the next section we therefore examine some of the issues associated with the industrial deployment of product intelligence.

3. PRODUCT INTELLIGENCE: PRACTICE

The deployment of product intelligence has been examined in a number of industrial domains such as manufacturing (McFarlane, 2003, Meyer, 2011), logistics (Kärkkäinen 2003, Meyer, 2011, Hülsmann, 2007, Giannikas, 2012) and services (Brintrup, 2011). We also comment on the numerous instances of more conventional systems displaying elements of product intelligence mentioned in Section 2.2 such as kanban and product recipe driven systems. The comments in the following sections refer in summary form to the applications described in these and other application oriented papers relating to product intelligence.

3.1 Deployment Strategies

We begin by examining different deployment options for product intelligence.

- *Fully / Partially Automated:* An intelligent product oriented process involves the completion fo the comtrol loop in Figure 5 – that is gathering sensed and other data relating to the product, decisions influencing the processing of the product or order and actuation or execution of the results of these decisions. An intelligent product approach would expect to automate some or all of these stages. Even a simple kanban system (see Slack, 2001) entails some automation of the decision making associated with triggering new production based on downstream demand, while logistics systems based on product intelligence have to date required human intervention in goods handling (Giannikas, 2012) and decision support (Meyer, 2011).

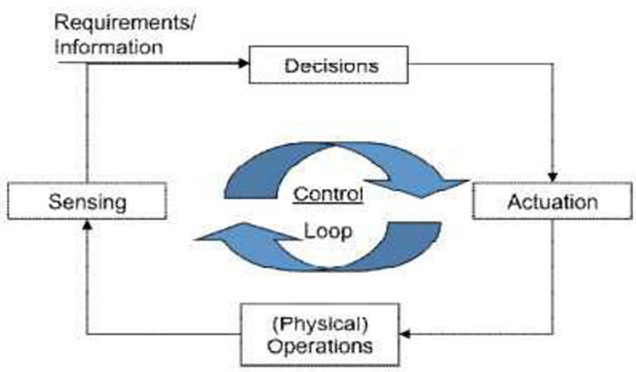

Fig. 5: Standard Feedback Control Loop

- *Partial / Full Pull Model:* This refers to whether the product intelligence mechanism is sufficient on its own to provide adequate operational control. A significant challenge for intelligent product based control is whether and how it must coexist with organisationally orented planning and control systems. We note that kanban systems are often deployed in conjunction with so called MRP or Material Requirements Planning systems which push supplied raw materials into a manufacturing environment on the basis of forecasts. In aircraft equipment servicing (Brintrup, 2011) the maintenance needs of an individual [intelligent] part must be balanced against the needs and operational schedules of the overall aircraft

- *Passive / Active Product Intelligence*: This distinction stems from Definition 3 in Section 2 and is also referred to as information vs decision oriented product intelligence. Here we distinguish between a product or order simply monitoring its progress and gathering appropriate information and directly influencing progress and potentially changing production, distribution, maintenance plans if conditions change. In product-driven manufacturing (McFarlane, 2003) changes to the customer order structure can directly influence manufacturing plans while in some supply chain management examples (Kärkkäinen, 2003, Kelepouris, 2011) product intelligence plays mainly a passive role focussing predominantly on accurate tracking and monitoring of order status.

- *On board / off board intelligence:* A key deployment decision is to what extent information and also potentially decision making capabilities are directly embedded on the physically order or product. At one extreme much of the RFID driven product intelligence models (Wong, 2003, Kärkkäinen, 2003) have considered a minimal style of deployment such as that illustrated in Fig 6 where a low memory RFID tag on the product is used as a pointer to information and decision processing on the internet. This has significant cost advantages although reliance on network access has meant that in other applications more memory capable RFID tags have been considered to support some local retention of information (Brintrup, 2011). Further, smart objects capable of local decision making at the product as well as information retention have been developed with applications for valuable items in mind (Lopez, 2011)

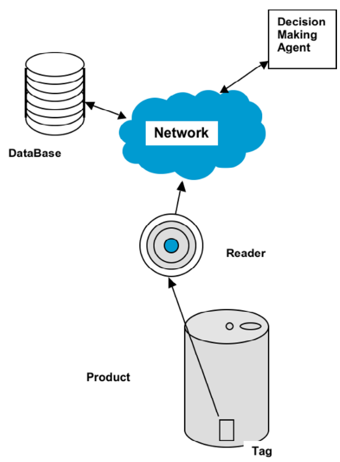

Fig 6: Off Board Product Intelligence Model

3.2 Enablers for Deploying Product Intelligence

In the companion paper to this paper (McFarlane, 2012), emerging business and technological enablers for deploying product intelligence have been discussed and these are summarised in Table 2.

Business Enablers	Technological Enablers
energy price constraints	*RFID Systems*
environmental constraints	*Object and Vehicle Location Systems*
tighter traceability regulations & practices	*Distributed Data Management Methods*
supply chain disruptions	*Order Tracking Software*
internet-based consumer services	*Web/Cloud Services*
multi-modal logistics	*Internet of Things*

Table 2: Business and Technological Enablers

The increased business motivation for product intelligence is – in the main part – created by the increasingly tough business environment for manufacturers, logistics providers and retailers in terms of constraints on energy consumption, carbon emissions, raw material shortages waste reduction and drug and food traceability legislations. All of these place a greater emnphasis on the accurate, efficient and well documented processing and transportation of goods – often at the level of individual orders. In logistics for example (Meyer, 2011, Giannikas, 2012) this can mean the need to manage information dynamically across multiple organisations and to consider changes to the modality used to ensure efficient transportation. If conditions change, direct access to customer intentions linked to orders can be of a significant benefit.

Technologically, major advances in location and identity sensing – RFID and GPS systems in particular – is revolutionising the ability to track orders and individual items accurately and multi-organisational item tracking and data management services allow an identified item to be linked to data held about it anywhere in its supply network. Web services can provide a customer with an increasingly rich interface to the operations of his supplier. In McFarlane (2012) it is observed that today's technological environment is much better equipped to support product intelligence compared to that of 10 years ago.

3.3 Barriers to Deploying Product Intelligence

We conclude this section by commenting on four classes of barriers to the deployment of intelligent products:

(a) *technical feasibility:* As discussed in Section 4.2, many of the building block for supporting product intelligence are either in place or under development. A major outstanding obstacle however is that of demonstrating that an intelligent product environment can be deployed with industrial scale information systems.

(b) *economic viability:* There is a significant gap in this area. While amost papers list potential advantages of the product intelligence approach more work is required in establishing quantifiable benefits to be gained through deployment of intelligent product based systems to justify commercial scale developments.

(c) *operational practicality:* Initial deployments are likely to be partial rather than full deployments (see Section 4.1) and hence validating the compatibility of any such development with other necessary information systems. In the longer term when active deployments are considered on a commercial scale the stability of operation of what are potentially highly distributed systems must be considered.

(d) *cultural acceptability:* A final barrier is gaining acceptability from the potential customers and users of the proposed systems. Product intelligence potentially implies a high level of transparency of supply operations to the ultimate customer which is likely to cause some initial discomfort.

4. CONCLUSIONS

This paper has outlined theoretical and practical issues associated with the development of the intelligent product paradigm and its industrial uptake. The path to product intelligence becoming part of today's industrial supply chain control environment needs to be considered cautiously. A migration approach in which capabilities are added to convention environments would appear provide a feasible route. In this way many of the difficulties in industrial deployment of agent based control and holonic manufacturing can be avoided.

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